

Lab Sheet 03 - Use Case Diagram

Group Report

1. Group Details

Group ID	2026-Y2-S1-MLB-WEB3G2-44
Topic Name	Web-based Recruitment Company System
Module	SE2030 Software Engineering
Lab	Lab 03 - Use Case Diagram

Group Members

No.	Name / Registration Number	Assigned Use Case (Section 4)
1	[Name] - [IT Number]	Apply for Job
2	[Name] - [IT Number]	Post Vacancy
3	[Name] - [IT Number]	Approve Job Posting
4	[Name] - [IT Number]	Approve Employer Registration
5	[Name] - [IT Number]	Submit Interview Feedback
6	[Name] - [IT Number]	Generate Invoice

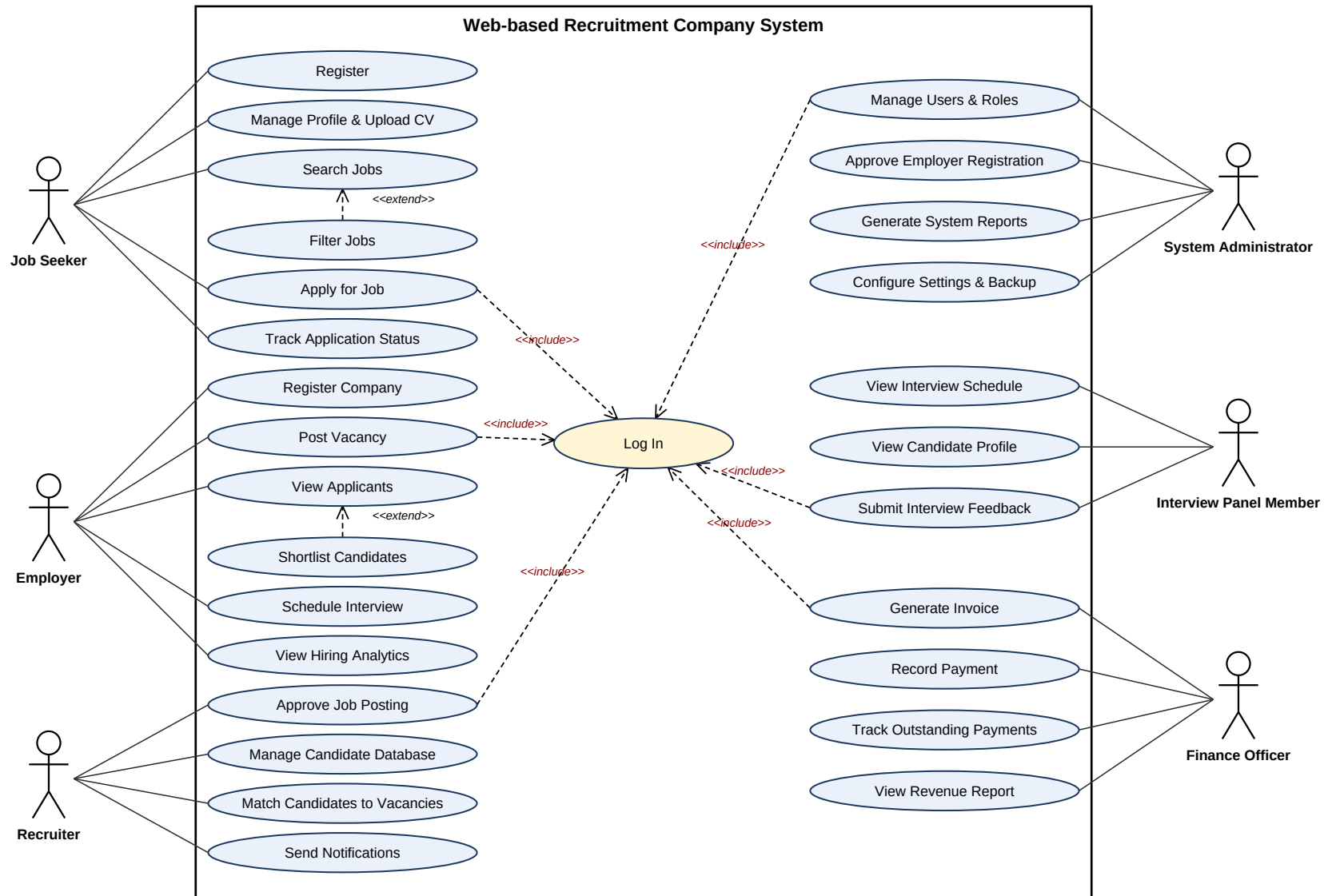
System Overview

The Web-based Recruitment Company System allows job seekers to build profiles and apply for jobs online, employers to post vacancies and select candidates, and recruiters to verify postings and match candidates. Administrators, interview panel members and finance officers support the process through user management, interview evaluation and billing of placement fees. This report models the system's functions and users with a UML use case diagram and describes one use case per member in detail.

Actors

Actor	Role in the System
Job Seeker	Registers, maintains a profile and CV, searches and applies for jobs, tracks applications.
Employer	Registers the company, posts vacancies, reviews and shortlists applicants, schedules interviews, views analytics.
Recruiter	Approves job postings, manages the candidate database, matches candidates to vacancies, sends notifications.
System Administrator	Manages users and roles, approves employer registrations, generates reports, configures settings and backups.
Interview Panel Member	Views interview schedules and candidate profiles, submits interview feedback.
Finance Officer	Generates invoices, records payments, tracks outstanding payments, views revenue reports.

2. Use Case Diagram



Solid lines: associations. Dashed arrows: <<include>> (all authenticated functions include **Log In**) and <<extend>> (optional behaviour). Editable version: UCD_2026-Y2-S1-MLB-WEB3G2-44.drawio (open in draw.io).

3. Use Case List

ID	Use Case	Primary Actor	Relationship
UC-01	Log In	All registered users	Included by other use cases
UC-02	Register	Job Seeker	
UC-03	Manage Profile & Upload CV	Job Seeker	
UC-04	Search Jobs	Job Seeker	
UC-05	Filter Jobs	Job Seeker	<<extend>> Search Jobs
UC-06	Apply for Job	Job Seeker	<<include>> Log In
UC-07	Track Application Status	Job Seeker	
UC-08	Register Company	Employer	
UC-09	Post Vacancy	Employer	<<include>> Log In
UC-10	View Applicants	Employer	
UC-11	Shortlist Candidates	Employer	<<extend>> View Applicants
UC-12	Schedule Interview	Employer	
UC-13	View Hiring Analytics	Employer	
UC-14	Approve Job Posting	Recruiter	<<include>> Log In
UC-15	Manage Candidate Database	Recruiter	
UC-16	Match Candidates to Vacancies	Recruiter	
UC-17	Send Notifications	Recruiter	
UC-18	Manage Users & Roles	System Administrator	<<include>> Log In
UC-19	Approve Employer Registration	System Administrator	
UC-20	Generate System Reports	System Administrator	
UC-21	Configure Settings & Backup	System Administrator	
UC-22	View Interview Schedule	Interview Panel Member	
UC-23	View Candidate Profile	Interview Panel Member	
UC-24	Submit Interview Feedback	Interview Panel Member	<<include>> Log In
UC-25	Generate Invoice	Finance Officer	<<include>> Log In
UC-26	Record Payment	Finance Officer	
UC-27	Track Outstanding Payments	Finance Officer	
UC-28	View Revenue Report	Finance Officer	

4. Individual Use Case Scenarios

Each member has written one comprehensive use case scenario for the function assigned to them.

4.1 UC-06: Apply for Job

Written by	Member 1 - [Name] - [IT Number]
Use Case ID / Name	UC-06 - Apply for Job
Primary Actor	Job Seeker
Supporting Actors	Employer (receives application), Email Service
Description	A registered job seeker applies online for a published vacancy using his/her stored profile and CV, so that the application reaches the employer and recruiter without any manual paperwork.
Preconditions	• The job seeker has a registered and active account and is logged in (includes Log In). • The job seeker's profile is complete and at least one CV is uploaded. • The selected vacancy is published and its closing date has not passed.
Trigger	The job seeker clicks Apply Now on a job details page.
Main Flow (Basic Path)	1. The job seeker opens the details page of a vacancy and selects Apply Now. 2. The system confirms the user is logged in and that the profile is complete. 3. The system displays the application form pre-filled with the profile details, the default CV and an optional cover letter box. 4. The job seeker reviews the details, optionally selects a different CV or uploads a new one, and writes a cover letter. 5. The job seeker selects Submit Application. 6. The system validates mandatory fields, CV file type/size, the closing date and that the user has not already applied. 7. The system saves the application with the status Submitted and a timestamp. 8. The system emails a confirmation to the job seeker and adds the application to the employer's and recruiter's applicant lists. 9. The system shows a confirmation page with an application reference number and a link to Track Application Status.
Alternative Flows	2a. User not logged in: system redirects to the login page and returns to the application after successful login. 2b. Profile incomplete: system prompts the user to complete the missing profile sections first. 5a. Job seeker cancels: system discards the form and returns to the job details page. 6a. Validation fails: system highlights the errors and keeps the entered data. 6b. Already applied: system informs the user and shows the existing application status. 6c. Vacancy closed or removed: system informs the user and disables the Apply button.
Exception Flows	• Upload or database failure: system shows an error message, keeps a draft of the form and lets the user retry. • Confirmation email fails: application is still saved; system queues the email for retry.
Postconditions	• The application is stored with status Submitted. • The employer and recruiter can view the application; the job seeker can track it.
Business Rules	• One application per job seeker per vacancy. • CV must be PDF or DOCX and not larger than 5 MB. • Applications are not accepted after the closing date.
Relationships	<<include>> Log In

4.2 UC-09: Post Vacancy

Written by	Member 2 - [Name] - [IT Number]
Use Case ID / Name	UC-09 - Post Vacancy
Primary Actor	Employer
Supporting Actors	Recruiter (approves the posting), Email Service
Description	An approved employer creates a new job vacancy so that it can be reviewed by the recruiter and published for job seekers.
Preconditions	• The employer's company account is approved and active. • The employer is logged in (includes Log In).
Trigger	The employer selects Post New Vacancy from the employer dashboard.
Main Flow (Basic Path)	1. The employer selects Post New Vacancy. 2. The system displays the vacancy form (job title, description, category, location, employment type, salary range, required skills, experience level, number of positions and closing date). 3. The employer fills in the details. 4. The employer selects Submit for Approval. 5. The system validates the mandatory fields and that the closing date is in the future. 6. The system saves the vacancy with the status Pending Approval. 7. The system notifies the recruiter that a new posting is waiting for review. 8. The system displays a confirmation message and the vacancy appears under My Vacancies as Pending.
Alternative Flows	4a. Employer selects Save as Draft: system saves the vacancy as Draft and it can be completed later. 5a. Validation fails: system shows error messages next to the invalid fields and keeps the data. 5b. Closing date in the past: system asks the employer to enter a valid date. 8a. Employer edits a pending vacancy: system updates it and keeps the status Pending Approval.
Exception Flows	• System or database error: system displays an error message and preserves the entered data as a draft. • Session timeout: system asks the employer to log in again and restores the draft.
Postconditions	• A vacancy exists with the status Pending Approval. • The vacancy is not visible to job seekers until approved.
Business Rules	• Only approved employers may post vacancies. • All mandatory fields must be completed before submission. • Every posting must be approved by a recruiter before publishing.
Relationships	<<include>> Log In

4.3 UC-14: Approve Job Posting

Written by	Member 3 - [Name] - [IT Number]
Use Case ID / Name	UC-14 - Approve Job Posting
Primary Actor	Recruiter
Supporting Actors	Employer (notified of decision), Email Service
Description	A recruiter reviews a vacancy submitted by an employer and approves, rejects or returns it for changes so that only genuine and appropriate vacancies are published.
Preconditions	• The recruiter is logged in (includes Log In). • At least one vacancy with the status Pending Approval exists.
Trigger	The recruiter opens the Pending Job Postings list.
Main Flow (Basic Path)	1. The recruiter opens the Pending Job Postings list. 2. The system lists pending vacancies with company, title and submitted date. 3. The recruiter selects a vacancy to review. 4. The system displays the full vacancy details and the employer's verification status. 5. The recruiter checks that the content is genuine, complete, non-discriminatory and not a duplicate. 6. The recruiter selects Approve. 7. The system changes the status to Published and makes the vacancy searchable. 8. The system records the action (recruiter, date and time) in the audit log. 9. The system emails the employer that the vacancy is live and displays a success message.
Alternative Flows	6a. Reject: the recruiter selects Reject and enters a reason; the status becomes Rejected and the employer is notified with the reason. 6b. Request changes: the recruiter enters comments; the status becomes Changes Requested and the employer is asked to edit and resubmit. 6c. Duplicate detected: the system warns the recruiter and offers to link or reject the posting.
Exception Flows	• Notification email fails: the decision is still saved and the system retries sending the email. • Concurrent review by another recruiter: the system informs the user that the posting has already been processed.
Postconditions	• The vacancy status is Published, Rejected or Changes Requested. • The decision is recorded in the audit log and the employer has been notified.
Business Rules	• Only users with the Recruiter role may approve postings. • A rejection or change request must include a reason. • Only approved vacancies appear in job search.
Relationships	<<include>> Log In

4.4 UC-19: Approve Employer Registration

Written by	Member 4 - [Name] - [IT Number]
Use Case ID / Name	UC-19 - Approve Employer Registration
Primary Actor	System Administrator
Supporting Actors	Employer (applicant), Email Service
Description	The system administrator verifies a company's registration request and approves or rejects it so that only genuine employers can use the recruitment system.
Preconditions	• The administrator is logged in (includes Log In). • An employer registration request with the status Pending exists.
Trigger	The administrator opens Employer Registration Requests .
Main Flow (Basic Path)	1. The administrator opens the Employer Registration Requests page. 2. The system displays the pending requests with company name, business registration number, contact person and date. 3. The administrator selects a request. 4. The system shows the full company details and uploaded documents. 5. The administrator verifies the details against the submitted documents. 6. The administrator selects Approve. 7. The system activates the employer account and generates a username and temporary password. 8. The system emails the credentials to the registered contact person. 9. The system logs the approval (administrator, date and time) and displays a success message.
Alternative Flows	6a. Reject: the administrator enters a reason; the status becomes Rejected and the applicant is notified. 6b. More information needed: the administrator requests additional documents; the status becomes On Hold and the applicant is notified. 4a. Duplicate company found: the system warns the administrator and links the existing account.
Exception Flows	• Credential email fails: the account stays active, the system flags the request and offers Resend Credentials. • Database error: the status is not changed and an error message is shown.
Postconditions	• The employer account is Active (approved) or the request is Rejected/On Hold. • The action is stored in the audit log and the applicant has been notified.
Business Rules	• Only administrators may approve employer registrations. • Temporary passwords must be changed at first login. • Every decision must be logged.
Relationships	<<include>> Log In

4.5 UC-24: Submit Interview Feedback

Written by	Member 5 - [Name] - [IT Number]
Use Case ID / Name	UC-24 - Submit Interview Feedback
Primary Actor	Interview Panel Member
Supporting Actors	Recruiter and Employer (view feedback), Email Service
Description	An interview panel member records scores and comments for a candidate after an interview so that hiring decisions are fair, consistent and documented.
Preconditions	• The panel member is logged in (includes Log In). • The panel member is assigned to the interview and the interview time has passed.
Trigger	The panel member selects a completed interview from My Interviews .
Main Flow (Basic Path)	1. The panel member opens My Interviews and selects a completed interview. 2. The system displays the candidate summary and the evaluation form. 3. The panel member rates each criterion (technical skills, communication, problem solving, cultural fit) from 1 to 5. 4. The panel member enters comments and selects a recommendation (Hire, Hold or Reject). 5. The panel member selects Submit Feedback. 6. The system validates that all mandatory ratings and the recommendation are provided. 7. The system saves the feedback and marks the interview as Feedback Completed. 8. The system notifies the recruiter and employer that feedback is available. 9. The system displays a confirmation message.
Alternative Flows	5a. Save draft: the system stores the form as a draft and it can be completed later. 2a. Candidate did not attend: the panel member marks No Show; scoring is skipped and the recruiter is notified. 6a. Validation fails: the system highlights the missing fields. 9a. Edit request: the panel member may edit the feedback within 24 hours of submission.
Exception Flows	• System error while saving: the entered data is preserved as a draft and the user is asked to retry. • Session timeout: the user logs in again and the draft is restored.
Postconditions	• Feedback is stored against the candidate's interview record. • The recruiter and employer can view the feedback.
Business Rules	• Only assigned panel members can submit feedback. • Submitted feedback is read-only after 24 hours. • Ratings are on a fixed 1-5 scale.
Relationships	<<include>> Log In

4.6 UC-25: Generate Invoice

Written by	Member 6 - [Name] - [IT Number]
Use Case ID / Name	UC-25 - Generate Invoice
Primary Actor	Finance Officer
Supporting Actors	Employer (receives invoice), Email Service
Description	A finance officer generates an invoice for the placement fee when a candidate has been successfully placed with an employer, so that the recruitment company can collect its income.
Preconditions	• The finance officer is logged in (includes Log In). • A placement with the status Confirmed exists and has not been invoiced. • The fee terms (percentage or fixed amount) are defined for the employer.
Trigger	The finance officer selects Placements Ready for Invoicing .
Main Flow (Basic Path)	1. The finance officer opens Placements Ready for Invoicing. 2. The system lists confirmed placements that have no invoice. 3. The finance officer selects a placement and chooses Generate Invoice. 4. The system retrieves the employer, candidate, vacancy and agreed salary details. 5. The system calculates the placement fee and taxes and creates a draft invoice with an invoice number, issue date and due date. 6. The finance officer reviews the draft and applies any approved adjustment or discount. 7. The finance officer selects Confirm and Issue. 8. The system generates the invoice as a PDF, stores it and sets the status to Issued. 9. The system emails the invoice to the employer and adds it to the outstanding payments list. 10. The system records the action in the audit log and displays a success message.
Alternative Flows	5a. Fee terms missing: the system asks the finance officer to enter the fee manually and records the change. 6a. Cancel: the finance officer discards the draft and the placement remains uninvoiced. 9a. Email delivery fails: the invoice stays Issued and the system offers Resend Invoice.
Exception Flows	• PDF generation fails: the system shows an error and keeps the draft so that it can be regenerated. • Invoice already exists for the placement: the system blocks duplicate creation and shows the existing invoice.
Postconditions	• An invoice with the status Issued is stored and linked to the placement. • The amount appears in the outstanding payments list.
Business Rules	• One invoice per confirmed placement. • Only users with the Finance Officer role can issue invoices. • Fee adjustments must be recorded in the audit log.
Relationships	<<include>> Log In